THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · NCERT-TYPE UNIT TEST SET 1

Chapters 1–4: **Exploring Substances: Acidic, Basic and Neutral, Electricity: Circuits and their Components, The World of Metals and Non-metals, Changes Around Us: Physical and Chemical** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. Substances that taste sour are generally: 1

(a) neutral **(b)** basic **(c)** salty **(d)** acidic

2. A device that converts chemical energy into electrical energy is: 1

(a) a switch **(b)** a fuse **(c)** an electric cell **(d)** a bulb

3. The shiny appearance of metals is called: 1

(a) sonority **(b)** malleability **(c)** ductility **(d)** lustre

4. A change in which no new substance is formed is a: 1

(a) physical change **(b)** chemical change

(c) biological change only **(d)** nuclear change

5. Which of these is a laboratory acid? 1

(a) sulphuric acid **(b)** citric acid **(c)** formic acid **(d)** lactic acid

6. Materials that allow electric current to pass through them are: 1

(a) insulators **(b)** cells **(c)** fuses **(d)** conductors

7. Non-metals are generally: 1

(a) sonorous **(b)** malleable **(c)** brittle **(d)** ductile

8. Which of the following is a reversible change? 1

(a) cooking egg **(b)** burning of candle

(c) rusting **(d)** melting of wax

9. Red litmus turns blue in: 1

(a) base **(b)** salt solution **(c)** acid **(d)** water

10. When current flows through a wire, the wire becomes: 1

(a) cold **(b)** hot **(c)** magnetic only **(d)** wet

11. When a metal burns in oxygen, it forms a: 1

(a) metal oxide **(b)** acid **(c)** salt **(d)** non-metal oxide

12. Rusting needs: 1

(a) air and moisture **(b)** only sunlight **(c)** only air **(d)** only water

13. Substances that feel soapy and taste bitter are generally: 1

(a) acidic **(b)** basic **(c)** sour **(d)** neutral

14. Two or more cells joined together form a: 1

(a) battery **(b)** switch **(c)** fuse **(d)** circuit breaker

15. The property of metals by which they can be beaten into thin sheets is: 1

(a) malleability **(b)** ductility **(c)** brittleness **(d)** sonority

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. What are indicators? Give two natural indicators. 2

17. What is an electric circuit? 2

18. Name four physical properties of metals. 2

19. What is a physical change? Give two examples. 2

20. Name two common bases used at home. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. Describe how you will test whether a solution is acidic, basic or neutral using litmus paper. 3

22. Explain why the bulb does not glow when the switch is off. 3

23. Describe the activity to test the conduction of electricity in metals and non-metals. 3

24. Classify as physical or chemical change: melting of wax, burning of wood, rusting, boiling of water. 3

OR

Why should we not taste unknown chemicals to test whether they are acids or bases? 3

25. How does a fuse protect an electrical appliance? 3

OR

Why is aluminium used for making foil and cooking vessels? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. Describe an activity to classify given household substances as acidic, basic or neutral using litmus, turmeric and China rose. 5

OR

Describe an activity to show that some materials conduct electricity and others do not. Name five objects. 5

27. Explain the physical properties of metals with an example of each. 5

OR

Describe the differences between physical and chemical changes with five examples each. 5

END OF QUESTION PAPER — Total: 50 Marks